Software Engineering Take-Home Assignment

Context

As AI capabilities advance rapidly, engineers at Anthropic face a unique challenge: building tools and experiences that are genuinely useful, trustworthy, and delightful. The best products demonstrate taste, handle complexity gracefully, and solve real problems in non-obvious ways.

This assignment asks you to build something that showcases your judgment, creativity, and execution.

Time Expectation

Target: 1-2 hours. Hard limit: 8 hours.

This assignment is designed to replace a synchronous interview session. We expect most candidates to complete something compelling in 1-2 hours, and we value your time - something insightful and deeper is more valuable than breadth. You may spend up to 8 hours if you're enjoying the problem, but this is not expected or required—we've seen excellent submissions completed in under 2 hours.

If you find yourself exceeding 8 hours, stop and submit what you have. Part of what we're evaluating is your ability to scope effectively.

In your written rationale, please note approximately how long you spent.

Objective

Build a functional prototype that demonstrates your ability to ship an impressive, self-contained experience. Choose one of the following themes:

Theme 1: Exploration & Understanding

Complex systems, technical concepts, and unfamiliar artifacts are hard to understand through static explanation. Build a tool that helps users develop deep understanding—whether that's a simulation of emergent dynamics, an explainer for a technical concept, or a tool for exploring codebases, datasets, or documents.

Theme 2: Creative & Generative Tools

Creative tools can unlock workflows that weren't previously possible. Build a tool that gives users meaningful creative leverage—not just "generate" buttons, but real control over iteration, variation, and refinement.

Theme 3: Systems & Reliability

Real systems must handle failure, concurrency, and scale. Build something that demonstrates thoughtful systems engineering—whether that's a developer tool that solves a real workflow pain point, a distributed primitive that handles failure gracefully, or infrastructure that degrades predictably under stress.

Theme 4: Evaluation & Data Quality

ML research depends on rigorous evaluation and clean data—but both are surprisingly hard. Build a tool or pipeline that addresses a real challenge in evals, reward modeling, data quality, or experiment tracking.

Theme 5: Performance Optimization

Performance engineering requires understanding hardware, algorithms, and tradeoffs. Build the fastest implementation you can of a meaningful component, with benchmarks demonstrating your results.

This theme is evaluated primarily on actual measured performance relative to baselines. Include benchmarks, methodology, and comparison to naive/reference implementations.

Critical Requirement: Self-Contained Evaluation

Your prototype must be evaluatable without requiring specific data, documents, or domain expertise from the reviewer.

If your concept requires domain-specific inputs, either bundle compelling examples or provide a "demo mode" that showcases the tool's capabilities without requiring the reviewer to source their own data.

Requirements

Create three deliverables:

1. A functioning prototype (deployed)

- The Anthropic team should be able to use and interact with the prototype immediately.
  - A browser or API-based experience is likely the simplest; a local experience should come with an appropriate way to run the project.
- Feel free to focus on a single feature or interaction pattern
- Polish is less important than demonstrating your core idea effectively
- Must include sample data or demo mode if the tool requires specific inputs

2. Your code

- Submit via GitHub repo link
- Code quality matters but is not the primary evaluation criterion
- Use any languages/technologies you're comfortable with

3. Design rationale

- Use both formats: self-recorded video (~5 min) and a short written doc
- Why you chose this theme and this specific approach
- What makes your idea interesting or non-obvious
- Key design decisions and tradeoffs you made

- How you'd extend this with more time
- Approximately how long you spent

Use of AI During Development

You are expected to use Claude Code, Claude.ai, or other AI tools during development. We want to see how you leverage these tools effectively—this is a core skill for engineers at Anthropic. You are permitted to use competitor models if you'd like.

Please submit your Claude/AI transcripts alongside your code. We're evaluating your judgment: how you direct the AI, evaluate its outputs, make tradeoffs, and maintain your own vision. A fully AI-generated project with no judgment will not pass.

Submission

Please submit via email:

- GitHub repo link with your code
- Link to your working prototype (hosted anywhere)
- Claude/AI transcripts showing your interaction history during development. You can generate these using a tool like [Claude Code transcripts] or the "Share" button on Claude.ai.
- Explanation artifacts (video + short written doc)